

RACING THE VIRUS

Making vaccines for India and the world

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Through the course of the pandemic, there were multiple announcements of tie ups with universities and research institutions from around the world, and some held up while others fell apart. While there are over 30 vaccines being developed in India, the major companies with vaccines in advanced stages of clinical trials are Serum Institute, Bharat Biotech, Biological E and Zydus Cadila. At the time of writing, Covishield and Covaxin are part

of India's vaccination program, with emergency approval given to Sputnik V. A number of pharmaceutical companies in India have announced partnerships to produce the vaccine locally, and the vaccine is expected to be the third major vaccine used in India's program. Close on its heels is another vaccine candidate by Serum Institute known as Covovax, which is in the advanced stages of clinical development. The most important thing to note here is that the whole world came together to accelerate the devel-

opment of vaccines. According to the initial projections at the time of the outbreak, the first vaccines were not expected to be available before 2022. The race to develop the vaccine began in earnest early last year. Serum Institute of India announced a series of partnerships and was conducting animal trials on two candidates when it announced the license for conducting the clinical trials of the Oxford vaccine. A major deal for delivery of 100 million doses was struck with Gavi, The Vaccine Alliance and the Bill & Melinda Gates Foundation.

Dr. Renu Swarup, Secretary, Department of Biotechnology, Government of India, said, "India has a proven track record of manufacturing safe and cost-effective vaccines not only for India, but for the world. Over the last decade, Government of India has encouraged innovation, and supported the Indian vaccine industry to manufacture high quality, affordable indigenous tools and products to benefit India and the world. We are very happy to see SII enter this global partnership to respond to the global health crisis posed by COVID-19." Dr. Balram Bhargava, Director General

KNOW YOUR VACCINES

Covishield

Manufacturer: Serum Institute

Developer: Oxford University and AstraZeneca

Tech: Viral vector



The vaccine cleverly uses a simian adenovirus as a trojan horse, with its own genes for replication switched off. It only introduces the genetic material that causes the cellular

machinery to manufacture the spike protein that the immune system learns to attack.

Covaxin

Manufacturer: Bharat Biotech

Developer: Bharat Biotech

Tech: Inactivated virus



The inactivated virus vaccines are more traditional forms of vaccination. The virus was isolated and allowed to replicate at scale using a vero cell manufacturing platform at Bharat

Biotech, after which they were inactivated using beta-propiolactone, that preserves their viral structures.

Sputnik V

Manufacturer: Dr. Reddy's and others

Developer: Gamaleya Research Institute

Tech: Viral vector



Instead of a single viral vector, the Sputnik V vaccine uses two adenoviruses with their replicative capabilities knocked off. Both contain the information necessary to manufacture the full length spike protein of SARS-CoV-2, to evoke the immune response from the body.

Covovax

Manufacturer: Serum Institute

Developer: Novavax and CEPI

Tech: Recombinant nanoparticle



A virus that infects moths is fed the genetic information to produce the SARS-CoV-2 spike protein and made to infect a culture of moth cells. The spike proteins are then harvested

and assembled onto a lipid nanoparticle, which is then injected into the body to train the system.

Bio E

Manufacturer: Biological E

Developer: Baylor College of Medicine and Dynavax Technologies



Tech: Protein subunit

The vaccine directly uses the spike proteins, and the Baylor College of Medicine was developing the platform since 2011, in anticipation of the next coronavirus outbreak. There is another vaccine candidate in the pipeline to address the genetic drift of the novel virus.

ZyCoV-D

Manufacturer: Zydus Cadila

Developer: Cadila Healthcare

Tech: DNA plasmid



ZyCoV-D use archaic organelles known as plasmids that are tiny self replicating capsules of DNA, which contain within them the genetic information needed for the cells to produce the viral proteins, which then elicits the immune response from the body and shields it from infection.